Diffusion Language Models Are Natively Length Aware

Vittorio Rossi[1*] Giacomo Cirò[1,2*] Davide Beltrame[1*] Luca Gandolfi[1,3] Paul Röttger[4,5] Dirk Hovy[1]

[1]Bocconi University [2]ETH Zurich [3]University of Zurich [4]Hasso Plattner Institute [5]University of Oxford *Equal contribution

A diffusion language model's first forward pass estimates the distribution of its answer length.

SmartCrop, which needs no training, cuts the canvas where the estimated length CDF first reaches 0.9. Against a fixed 1024-token canvas it decodes **1.05–2.4× faster** with threshold decoding and **1.4–6× faster** with one token per step, and loses at most 3.8 points of accuracy.

The padding tax

A diffusion LM denoises a fixed-length canvas, and every step runs over all of it, end-of-sequence (EoS) padding included.



Share of the 1024 canvas positions, LLaDA-8B on GSM8K

EoS probability = length CDF

Under EoS padding, with prompt c, canvas z_t at noise level t and B the first EoS position, the loss-optimal $p(\text{EoS})$ at masked position i is the probability that the answer has ended by i,

$$p^*(\text{EoS} \mid c, z_t, i) \;=\; \Pr(B \le i \mid c, z_t).$$

At $t = 1$ all positions are masked, so a trained model's $p(\text{EoS})$ from one pass estimates the CDF of B given the prompt and canvas length.

SmartCrop



1. **Read** $p(\text{EoS})$ in one pass at $t = 1$.
2. **Crop** where $p(\text{EoS})$ first reaches q.
3. **Denoise** the shorter canvas as usual.

No training. The extra pass takes 58 ms on an H100.

Limits

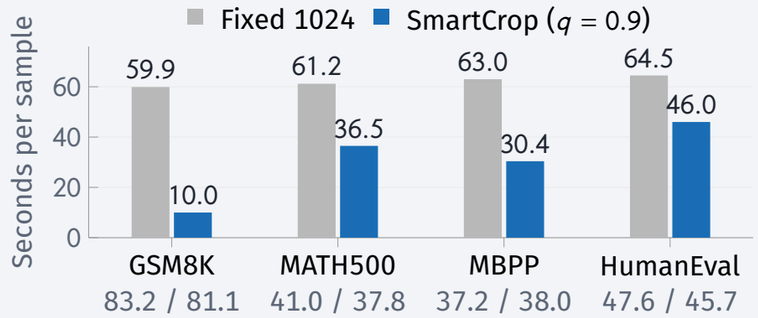
The reported $q = 0.9$ was chosen on test sets; it covers 78% of sampled GSM8K response lengths.

A task-mean fixed canvas has no significant accuracy difference in 7 of 8 model–task pairs.

Math and code only; single runs at batch size 1.

Faster, at comparable accuracy

LLaDA-8B-Instruct, **one token per step**. Accuracy (%) below each benchmark, fixed / SmartCrop.



Threshold decoding: 1.05–2.4× faster

Seconds per sample; accuracy (%) in grey. Confidence above 0.9; DAEDAL and ρ-EoS start at 64 tokens.

	GSM8K	MATH500	MBPP	HumanEval
Fixed 1024	7.0 83.1	12.0 40.2	8.8 37.0	12.0 46.3
DAEDAL	7.2 83.6	20.9 39.8	14.3 40.8	22.3 46.3
ρ-EoS	4.9 84.0	12.2 35.2	10.9 40.8	19.2 48.8
SmartCrop	**2.9** 82.2	**9.9** 36.4	**6.6** 37.2	**11.4** 47.0